

1. DEFINITION

DEFINITION
Glycerol distearate (type I) EP / Glyceryl distearate NF

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	Glyceryl Distearate
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	73071MW2KM (Glyceryl Distearate)
ADDITIONAL INFORMATION ON THE DENOMINATION	Glycerol esters of fatty acids Distearin
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE
This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (oral, topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Esterification reaction between Glycerol and Stearic acid 50. The esterification step is followed by atomization using a spray-cooling technique. No solvent or catalyst used during the manufacturing process.

COMPOSITION	
Well defined multi-constituent substance composed of mono-, di- and triglycerides of Palmitostearic acids (C16 and C18), the diester fraction being predominant (Specification: 40-60%).	
Full Labeling (Key letters system)	
A ₁ ≥ 75.0 ; 50.0 ≤ A ₂ < 75.0 ; 25.0 ≤ B < 50.0 ; 10.0 ≤ C < 25.0 ; 5.0 ≤ D < 10.0 ; 1.0 ≤ E < 5.0 ; 0.1 ≤ F < 1.0 ; G < 0.1	
Glyceryl Distearate	A ₁ ≥ 75.0%
Other constituents (i.e. incidental ingredients)	
None	



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4. RSPO CERTIFICATION

ROUNDTABLE ON SUSTAINABLE PALM OIL http://www.rspo.org/certification	
Product status	Gattefossé RSPO certification number
Product certified RSPO Mass Balance. More information on: www.RSPO.org	Supply Chain Certificate (SCC) number: BVC-RSPO-1-1106949778

5. DRUG MASTER FILE

Country	Master file number
US DMF (Type IV – Excipient)	5116
Canada MF (Type III – Excipient)	/
For submission of the DMF in another country, please contact your local Gattefossé representative	

6. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	Conforms to monograph
European Pharmacopoeia (Ph.Eur.) Current edition	Glycerol distearate (type I)
USP/ NF Current edition	Glyceryl distearate

7. VETERINARY MEDICINES FOR FOOD PRODUCING ANIMALS

EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
✓	⊗ ⁽¹⁾	✓	✓ ⁽⁵⁾⁽⁶⁾	✓ ⁽⁸⁾ under CAS 1323-83-7	✓ ⁽⁸⁾ 40CFR 180.910 under CAS 1323-83-7



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EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
EUROPEAN REGULATION (1) Regulation (EU) No 37/2010 - Listed in annex II (Food additives: substances with a valid E number approved as additives in food stuffs for human consumption) (2) Regulation (EU) No 37/2010 - Listed in annex II (Polyethylene glycol stearates with 8-40 oxyethylene units) (3) Regulation (EU) No 37/2010 - Listed in annex II (Diethylene glycol monoethyl ether) (4) EMA/CVMP - Substances considered as not falling within the scope of Regulation (EC) No 470/2009, with regard to residues of veterinary medicinal products in foodstuffs of animal origin USA REGULATION - Veterinary drugs FDA is responsible for assuring that animal drugs are not only safe and effective for animals, but that food products from treated animals are safe for humans to consume. In the case of a drug for food producing animals, the inactive ingredients of the drug or its composition may be safe with respect to human food safety (5) Approved food additives and GRAS substances for human food - Title 21, Part 172, 182–184 of the Code of Federal Regulations (CFR) (6) Approved excipients (inactive ingredients) for use in human drugs by oral route (Ref: FDA Inactive ingredient database) - Pesticides (8) EPA's pesticides program (http://iaspub.epa.gov/apex/pesticides/f?p=INERTFINDER:1:0::NO:1::)					

8. SPECIFIC PHARMACEUTICAL REGULATIONS

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	Ingredient Name	Use	Restrictions
Approved	Glyceryl distearate	Available for use as an Active Ingredient in: Biologicals, Prescription Medicines Available for use as an Excipient Ingredient in: Biologicals, Devices, Export Only, Listed Medicines, Over the Counter, Prescription Medicines Not available as an Equivalent Ingredient in any application	Topical Use Only (Export Only, Over the Counter, Prescription Medicines) Only for use in topical medicines for dermal application (listed medicines)
Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homoeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA NHPID		
Status	NHPID name	Route of administration/ Restrictions
Approved chemical name	Glyceryl distearate	Role Non-medicinal: Purposes: Lubricant , Skin-Conditioning Agent - Emollient Route Of Administration: Topical Allowed Purposes: Skin-Conditioning Agent - Emollient Detail: Up to 7%. Route Of Administration: Oral Allowed Purposes: Lubricant



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REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA [NHPID](#)

Status	NHPID name	Route of administration/ Restrictions
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		

9. SAFETY ASSESSMENT

SAFETY STUDIES PERFORMED BY GATTEFOSSE

Study type/ duration	Route	Species	Test article	Results/Conclusions	Reference
Acute cutaneous irritation and corrosion Test (OECD n°404)	Dermal	Rabbit	In liquid paraffin	Erythema (mean): 0.5 Oedema (mean): 0.00	Mildly Irritant Not classified
Acute Ocular Irritation and Corrosion test (OECD n°405)	Ocular	Rabbit	No dilution	/	Slightly Irritant Not classified
Acute cutaneous irritation test (Single patch test)	Dermal	Human	10% in stone oil	/	Non irritant Not classified
Acute oral Toxicity	Oral	Rat	Vehicle CMC 1%	LD ₀ ≥ 6000 mg/kg/day	Not classified
Acute oral Toxicity—Limit test (OECD n°401)	Oral	Rat	Olive oil	LD ₅₀ ≥ 2000 mg/kg/day	Not classified

CIR COMPENDIUM (Cosmetic Ingredient Review) (<http://www.cir-safety.org>)

Denomination	Reference	Maximum “as used” concentration	Conclusions
Glyceryl distearate	IJT 26(Suppl. 3):1-30, 2007	Up to 7.0 %	The CIR Expert Panel concluded that Glyceryl Diesters described in the assessment are safe as cosmetic ingredients in the practices of use and concentrations as described in this safety assessment provided that the content of 1,2-diester is not high enough to induce epidermal hyperplasia.



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10. CHEMICAL REGULATIONS

CHEMICAL REGULATION			
	CAS #	EC #	Chemical name
Glyceryl Distearate	85251-77-0 (or 1323-83-7)	286-490-9 (or 215-359-0)	Glycerides, C16-18 mono- and di- (or Glyceryl distearate)
EC numbers: There are three separate inventories. These are the European Inventory of Existing Commercial Chemical Substances (EINECS), the European List of Notified Chemical Substances (ELINCS) and the No-Longer Polymers (NLP) list. EINECS numbers start with 2 or 3. ELINCS numbers start with 4. NLP numbers start with 5. Substances without EC numbers were assigned "list numbers" in order to facilitate REACH submissions. They are solely of administrative, and not of regulatory relevance. The list numbers start with 6, 7, 8 or 9.			

NATIONAL INVENTORIES					
	U.S.A. (TSCA)	Canada (DSL/NDSL)	Australia (AIIC)	China (IECSC)	Other inventories
Glyceryl Distearate	Listed	DSL listed	Listed	Listed	Glycerides, C16-18 mono- and di-: AIIC, DSL, EINECS, NZIoC, PICCS, REACH, TCSI, TSCA, VNECI, AREC, ECL / Glycerides, C16-18 mono- and di-ester: IECSC (or) Octadecanoic acid, diester with 1,2,3-propanetriol: AIIC, DSL, ENCS, IECSC, NZIoC, PICCS, TCSI, TSCA / distearic acid, diester with glycerol: EINECS, PICCS, REACH / Octadecanoic acid diester with 1,2,3-propanetriol: AREC, ECL
U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics... are not subject to TSCA regulation. "Naturally occurring substances" are not subject to Premanufacture Notice reporting. Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation. Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities. Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration. Chemicals not listed on the AIIC must be assessed before importation.					

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
	Status of the product	Status per substance
Glyceryl Distearate	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACH regulation.



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11. HISTORY OF USE

11.1. Pharmaceutical applications

11.1.1. FDA IIG reference for Chemical equivalent

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
GLYCERYL DISTEARATE (UNII : 73071MW2KM)	ORAL	CAPSULE, EXTENDED RELEASE	39.2 mg	
	ORAL	TABLET		100 mg
	ORAL	TABLET, EXTENDED RELEASE		13 mg
	ORAL	TABLET, ORALLY DISINTEGRATING	4 mg	
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

11.1.1. FDA IIG references to the most similar chemical (read-across approach)

In the **read-across approach**, endpoint information for one chemical (the source chemical) is used to predict the same endpoint for another chemical (the target chemical), which is considered to be "similar" in some way (usually on the basis of structural similarity or on the basis of the same mode or mechanisms of action). In principle, read-across approach can be used to assess physicochemical properties, toxicity, environmental fate and ecotoxicity.

For any of these endpoints, read-across approach may be performed in a qualitative or quantitative manner.'

The similarities may be based on the following:

- a common functional group (e.g. aldehyde, epoxide, ester, specific metal ion);
- common constituents or chemical classes, similar carbon range numbers;
- an incremental and constant change across the category (e.g. a chain-length category);
- the likelihood of common precursors and/or breakdown products, via physical or biological processes, which result in structurally similar chemicals.

These similarities justify data gap filling in a chemical category by applying read-across.

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
GLYCERYL MONO AND DIPALMITOSTEARATE (UNII: KC98R082HJ)	ORAL	CAPSULE		16 mg
	ORAL	CAPSULE, DELAYED RELEASE		25 mg
	ORAL	TABLET, DELAYED RELEASE		3 mg
	ORAL	TABLET, ORALLY DISINTEGRATING, DELAYED RELEASE		28 mg



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FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
	TOPICAL	CREAM		312 mg
	VAGINAL	CREAM		160 mg
GLYCERYL PALMITOSTEARATE (UNII : GSY510183C)	INTRADERMAL	INJECTION	16 mg	
	ORAL	SUSPENSION	150 mg/ 5.00 ml	
	ORAL	TABLET	18 mg	
	TOPICAL	AEROSOL	2.55 % w/w	
	TOPICAL	CREAM	2.78 % w/w	
	TOPICAL	EMULSION	2.55 % w/w	
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

11.1.2. Other references

Glyceryl Monostearate and Glyceryl Palmitostearate are listed in the handbook of pharmaceutical excipient and are similar compounds that are identified using the read-across approach.

HANDBOOK OF PHARMACEUTICAL EXCIPIENT (Current edition by Paul J. Sheskey, Bruno C Hancock, Gary P Moss Dabid J Goldfard)			
Denomination	Functional Category	Regulatory status	Safety
Glyceryl Monostearate	Emollient Emulsifying agent Emulsion Stabilizing agent Modified-release agent Solubilizing agent Tablet and Capsule lubricant	GRAS listed. Included in the FDA Inactive Ingredients Database. Included in nonparenteral medicines licensed in the UK. Included in the Substance Added to Food (Formerly EAFUS) list compiled by the FDA. Included in the Canadian Natural Health Products ingredients Database.	Glyceryl monostearate is widely used in cosmetics, foods, and oral and topical pharmaceutical formulations, and is generally regarded as a nontoxic and nonirritant material. LD50 (mouse, IP) : 0.2 g/kg
Glyceryl Palmitostearate	Biodegradable material; coating agent; gelling agent; modified-release agent; tablet and capsule diluent; tablet and capsule lubricant; taste-masking agent.	GRAS listed. Included in the FDA Inactive Ingredients Database. Included in the Substance Added to Food (Formerly EAFUS) list compiled by the FDA. Included in the Canadian Natural Health Products ingredients Database. Included in nonparenteral preparations licensed in the UK.	Glyceryl palmitostearate is used in oral pharmaceutical formulations and is generally regarded as an essentially nontoxic and nonirritant material. LD50 (rat, oral) : > 6 g/kg



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COMPENDIUM OF PHARMACEUTICAL EXCIPIENTS FOR VAGINAL FORMULATIONS		
Denomination	Category (concentration in %)	regulatory status
Glyceryl monostearate	emulsifier; stabilizer; emollient; plasticizer; tableting aid	G, 1, 3, 8–10, 12–14, 23, 25, 26
Glyceryl stearate	emulsifying agent; stabilizer; emollient; plasticizer; tablet lubricant; sustained-release tablets	G, 12, 25
Glycerides of fatty acids	Suppository bases	1, 3–14, 16–26
Abbreviations: G: GRAS, I: inactive ingredients guide, Pharmacopeias, 1: Austrian, 2: Belgian, 3: British, 3v: British (veterinary), 4: Brazilian, 5: Chinese, 6: former Czechoslovakian, 7: Egyptian, 8: European, 9: French, 10: German, 11: Greek, 12: Hungarian, 13: Indian, 14: Italian, 15: International, 16: Japanese, 17: Mexican, 18: Netherlands, 19: Nordic, 20: Portuguese, 21: Romanian, 22: Russian, 23: Swiss, 24: Turkish, 25: United States, 26: former Yugoslavian Garg et al., <i>Pharmaceutical Technology, Drug Delivery</i>, 2001, p14–24, Compendium Of Pharmaceutical Excipients For Vaginal Formulations		

11.2. Food/Nutraceutical/Dietary supplement applications

Glyceryl esters of fatty acids have a long history of use worldwide as food additives for use in food, nutraceutical products and dietary supplements.

Country	Reference	Registration number
International	Evaluations Performed by Joint FAO/WHO Expert Committee on Food Additives	INS471 Mono- and Diglycerides
E.U.	EFA (European Food additives)	E471 Mono- and Diglycerides
U.S.A.	FCC (Food Chemical Codex)	Mono- and Diglycerides
U.S.A.	FCC (Food Chemical Codex)	Glyceryl palmitostearate
U.S.A.	GRAS US Food additives according to 21 CFR	21CFR §184.1505 Mono- and Diglycerides
U.S.A.	GRAS Generally recognized as safe according to 21 CFR	21CFR §184.1329 Glyceryl Palmitostearate
JAPAN	JSFA (Japanese Standards of Food Additives)	Glycerol esters of fatty acids

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